A Hybrid Multi-Criteria Decision Making and Ensemble Learning Framework for Performance Assessment: Evidence from Vietnam's PAPI

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Abstract

This study develops a hybrid framework for sub-national performance assessment, integrating hierarchical object-weighting mechanisms with a stacking ensemble meta-learning architecture. Utilizing the Vietnam Provincial Governance and Public Administration Index (PAPI) dataset as a longitudinal case study (2011–2024), we propose a three-pillar methodology designed to ensure robust, objective, and predictive indexing. The framework first employs a regime-aware, two-level CRITIC weighting procedure to derive objective importance coefficients that account for both contrast intensity and inter-criteria redundancy. Second, we establish a robust consensus ranking through an ensemble of five distinct multi-criteria decision-making (MCDM) algorithms (TOPSIS, VIKOR, PROMETHEE II, COPRAS, and EDAS), mitigating methodological idiosyncrasies. Third, we implement a production-hardened stacking ensemble comprising CatBoost, Bayesian Ridge, Support Vector Regression, and ElasticNet models to forecast provincial governance trajectories. Uncertainty is rigorously quantified through distribution-free conformal prediction intervals, providing 95% valid coverage for future governance outcomes. Our results demonstrate that the hybrid framework significantly outperforms traditional linear indexing approaches, offering precise insights into provincial governance efficiency and a forward-looking tool for evidence-based policy reform.

Keywords: Multi-Criteria Decision Making; CRITIC; Ensemble Learning; Stacking; Conformal Prediction; Performance Assessment; PAPI; Vietnam

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1 Introduction

The empirical assessment of sub-national governance capacity has emerged as a critical frontier in development economics and public administration research. Effective governance—operationalized through dimensions of transparency, accountability, administrative efficiency, and service delivery quality—constitutes a fundamental determinant of economic growth, institutional legitimacy, and citizen welfare across developing economies (**Acemoglu2012**; **Kaufmann2011**). Nowhere is this imperative more acute than in Vietnam, where the transition from a centrally planned to a market-oriented economy has necessitated substantive institutional reforms at provincial and municipal levels, yet remains hampered by inconsistent governance performance across the 63 administrative divisions (**Abramo2018**; Centre for Community Support and Development Studies (CECODES) et al., 2024).

Over the past fifteen years, multiple governance assessment frameworks have been institutionalized within Vietnam to benchmark provincial performance and catalyze administrative reform. The Provincial Governance and Public Administration Performance Index (PAPI), administered jointly by CECODES, the Vietnam Fatherland Front Research and Training Committee, and the United Nations Development Programme, has established itself as the preeminent system for measuring citizen-perceived governance quality across eight hierarchically-organized dimensions—civic participation, transparency, accountability, anti-corruption efficacy, administrative procedural efficiency, public service delivery quality, environmental governance, and e-governance accessibility (Centre for Community Support and Development Studies (CECODES) et al., 2024). Parallel frameworks interrogate complementary dimensions: the Satisfaction Index of Public Administrative Services (SIPAS) focuses on citizen satisfaction with bureaucratic transactions; the Provincial Competitiveness Index (PCI) combines governance metrics with economic and business-environment indicators; the Digital Transformation Index (DTI) assesses technological adoption and digital service maturity; and the Public Administration Reform Index (PAR Index) tracks incremental progress in administrative modernization (**VCCI2023**; **Tran2020**). Collectively, these indices constitute a rich empirical substrate for comparative provincial governance assessment, yet their methodological foundations remain fragmented, frequently reliant on ad hoc aggregation choices rather than principled, data-driven approaches (**Arndt2016**; **Arndt2012**).

The dominant methodological consensus in governance indexing practice—both in Vietnam and globally—has gravitated toward composite index construction via equal weighting of constituent indicators. This paradigm assumes that all dimensions contribute equally to the composite governance construct, assigning each indicator a symmetric $1/n$ normalizing weight. While equal weighting offers transparent communicability and computational simplicity, it confronts a critical limitation: equal weighting discards two vital sources of empirical information embedded in the data itself. First, indicators exhibit heterogeneous *discriminatory power*, or contrast intensity, across provinces. An administrative procedure indicator that yields identical scores across all 63 provinces carries zero ranking information,

yet under equal weighting receives the same axiological weight as a corruption perception indicator that substantially differentiates provinces. Second, indicators exhibit heterogeneous *redundancy*, or information overlap. When multiple indicators measuring related aspects of governance (e.g., access-to-information versus transparency-of-budget-execution) correlate strongly, equal weighting implies that correlated information is double-counted, inflating its influence in the final composite ranking (Diakoulaki et al., 1995; Greco et al., 2016).

Beyond weighting methodology, existing Vietnamese governance indices have adopted singular ranking aggregation methods. The PAPI composite, for instance, employs simple additive aggregation of normalized component scores after equal weighting. The PCI relies on principal component analysis for dimension reduction, followed by ad hoc element weighting. This reliance on a single aggregation paradigm introduces an underappreciated form of methodological fragility: the methodological idiosyncrasy problem. A fundamental theorem in multi-criteria aggregation theory establishes that no single aggregation function satisfies all desirable axiomatic properties simultaneously (**Arrow1951**; **Bouyssou2000**). Distance-based methods (e.g., Technique for Order of Preference by Similarity to Ideal Solution, TOPSIS) minimize deviations from an ideal solution profile but ignore pairwise outranking relationships. Outranking methods (e.g., VIKOR, Preference Ranking Organization Method for Enrichment Evaluation) model preference-based dominance but may introduce incomparability regions or intransitivity. Proportional assessment methods (e.g., Weighted Sum Method, COPRAS) assume cardinal utility linearity, a strong assumption violated when governance dimensions exhibit non-linear interactions. Consequently, reliance on any single method risks ranking instability when the underlying preference model is altered even marginally (**Wang2009**; Greco et al., 2016).

Despite the maturity attained by Vietnam's governance indices over nearly 14 years of annual measurement (2011–2024), three methodological gaps remain insufficiently addressed in extant applied research and policy practice.

Gap 1: Absence of principled, data-driven weighting schemes validated against temporal stability and robustness criteria. Current Vietnamese indices employ either equal weighting (canonical yet informationally wasteful) or subjective expert weighting (transparent in rationale but vulnerable to *expert bias* and unstable across different panels). No published study systematicically applies objective weighting methods—such as the Criteria Importance Through Intercriteria Correlation (CRITIC) method—to the hierarchical structure of the PAPI or its peer indices, nor does extant literature validate whether estimated weights exhibit temporal stability across the full measurement window (2011–2024), or remain robust to methodological perturbations. This gap is non-trivial: if governance priorities shift substantially over 14 years, a single unified weight vector may conceal important structural breaks; conversely, if weights prove unstable, confidence in the underlying governance assessment methodology itself is undermined. Formal temporal validation and sensitivity analysis remain conspicuously absent from the governance indexing literature on Vietnam.

Gap 2: Absence of multi-methodological consensus ranking frameworks that achieve robustness across diverse MCDM algorithms. The architectural assumption implicit in single-method ranking—that a particular mathematical formalism (distance minimization, outranking logic, or proportional aggregation) correctly encodes the governance assessment problem—lacks empirical validation. No published study applies a multi-method ensemble of MCDM algorithms to Vietnamese governance data, nor does extant research characterize the sensitivity of provincial rankings to methodological variation. This omission is significant because practitioners and policymakers require confidence that provincial governance orderings reflect genuine performance differentiation, not ephemeral artifacts of mathematical choice. A consensual ranking framework incorporating multiple methods with transparent aggregation rules would substantially elevate methodological credibility and policy robustness.

Gap 3: Disconnection between static MCDM ranking assessment and dynamic temporal forecasting of future governance trajectories. Existing governance indices are inherently retrospective: they rank provinces based on observed data from a single calendar year or multi-year average, without explicitly projecting likely governance evolution. For forward-looking governance planning and resource allocation, policymakers require not only current rankings but also probabilistic forecasts of future performance. Machine learning ensemble forecasting—which combines diverse base learners with metalearned combination weights—constitutes a methodologically mature framework for temporal prediction. Yet no study integrates hierarchical MCDM, temporal weighting validation, and ensemble machine learning forecasting into a unified governance assessment architecture. The absence of this integration means that governance assessment remains siloed from the powerful arsenal of time-series and ensemble learning methodologies that have transformed demand forecasting, economic nowcasting, and other high-stakes prediction domains (**Hastie2009**; Wolpert, 1992).

To address these three methodological gaps simultaneously, the present study develops and validates a comprehensive hybrid **MCDM-Ensemble Machine Learning Framework** for governance assessment, applied to the PAPI dataset spanning 14 years (2011–2024) and 63 Vietnamese provinces. The framework synthesizes three complementary innovations.

Innovation 1: Hierarchical CRITIC Weighting with Temporal Stability Validation. We extend the classical CRITIC method—which derives criterion weights from data heterogeneity (variance) and inter-criterion independence (correlation)—to a novel two-level hierarchical architecture tailored to the multi-tier structure of the PAPI dataset. Level 1 applies CRITIC independently within each of the 8 governance criteria, computing data-driven weights for the 29 constituent sub-criteria while respecting hierarchy and preventing over-weighting of sub-criteria in weak-signal criteria. Level 2 applies CRITIC across the 8 criterion-level composite scores, reflecting both within-criterion redundancy mitigation (Level 1) and inter-criterion information assessment. Critically, we conduct multi-window temporal stability analysis using overlapping 5-year windows, Spearman's rank

correlation, Kendall's omnibus concordance coefficient W, and per-criterion coefficient-of-variation metrics. This analysis empirically establishes whether governance prioritization—operationalized through the weight vector—exhibits sufficient temporal consistency across the 14-year span to justify deployment as a unified parameter, or whether time-variant weighting would be warranted. Additionally, three-tier Monte Carlo perturbation sensitivity analysis (conservative $\pm 5\%$, moderate $\pm 15\%$, aggressive $\pm 50\%$) quantifies ranking robustness to weight specification uncertainty, addressing a critical but under-examined aspect of indexing methodology.

Innovation 2: Multi-Methodological MCDM Consensus Ranking Framework. Rather than rely on a single ranking algorithm, we employ an ensemble of five distinct MCDM methods representing three orthogonal aggregation paradigms: (i) distance-based (TOPSIS), (ii) outranking (VIKOR, Preference Ranking Organization Method for Enrichment Evaluation), and (iii) proportional assessment (Weighted Sum Method, COPRAS, Evaluation Based on Distance from Average Solution). For each method and each year (2011–2024), independent provincial rankings are generated. Rank-order correlations and agreement indices (Spearman's ρ, Kendall's tau-b) characterize inter-method concordance, identifying methodological consensus regions and divergence zones. Final provincial rankings are derived via weighted aggregation of method-specific rankings, with method weights determined through bootstrap resampling-based stability analysis. This multi-method architecture explicitly acknowledges that no single mathematical formalism fully captures the governance assessment problem, and that consensus across diverse methods substantially elevates confidence in the robustness of provincial performance orderings.

Innovation 3: Integration with Ensemble Machine Learning Forecasting. The validated weighting scheme and multi-method ranking framework are integrated into an ensemble forecasting pipeline that projects likely governance trajectories (2025–2029). An ensemble of heterogeneous base learners—including regularized ridge regression, gradient boosting (CatBoost, XGBoost), kernel methods (Kernel Ridge Regression), support vector regression, quantile forest regression for interval prediction, and Bayesian additive regression trees—are trained on the temporally-layered panel dataset, using hierarchical CRITIC weights as auxiliary features signaling data quality and temporal dynamics. Meta-learner combination weights are estimated via non-negative least squares, ensuring interpretability and theoretical soundness. Out-of-fold prediction performance is characterized through multiple metrics (mean absolute error, continuous ranked probability score, quantile loss), and prediction intervals are derived via conformal prediction methodology, providing probabilistic governance forecasts aligned with natural forecast uncertainty.

The full framework is implemented in a reproducible, modular Python software architecture, enabling practitioners and policymakers to apply the approach to Vietnam's governance indices or parallel assessment frameworks. A rigorous comparative validation demonstrates that the hierarchical CRITIC-weighted, multi-method MCDM approach yields rankings that are temporally stable, methodologically robust, and substantially concordant across di-

verse aggregation algorithms—improvements over existing practice.

Roadmap of the paper: Section 2 describes the data, the hierarchical CRITIC weighting procedure with temporal stability and sensitivity analysis, and the multi-method MCDM framework. Section 3 presents empirical results characterizing Vietnam's provincial governance landscape, temporal weight evolution, and consensus versus divergence across MCDM methods. Section 4 integrates the validated weighting scheme into an ensemble forecasting pipeline and projects likely governance trajectories through 2029. Section 5 synthesizes findings, contextualizes contributions relative to extant governance indexing methodologies, discusses practical implications for Vietnamese policymakers, and identifies directions for future research.

2 Methods and Data

2.1 Data Description

The Viet Nam Provincial Governance and Public Administration Performance Index (PAPI) constitutes a comprehensive sociological surveillance instrument designed to evaluate the efficacy of state performance and public service delivery through the lived experiences of the Vietnamese citizenry. Methodologically grounded in representative household surveys, the index provides a multi-dimensional assessment of sub-national governance quality, encompassing critical domains such as transparency, vertical accountability, and the control of corruption (Centre for Community Support and Development Studies (CECODES) et al., 2024). Since its inception, PAPI has evolved into a pivotal evidentiary roadmap for provincial authorities, offering standardized metrics to benchmark institutional progress and catalyze systemic administrative improvements within the national development context.

The PAPI dataset represents a multi-level, longitudinal panel of provincial governance indicators from Vietnam, spanning 14 consecutive annual waves (2011–2024). This study structures the data as a hierarchical system consisting of three analytical levels: (i) a temporal dimension indexed by survey year t, (ii) a spatial dimension indexed by province i, and (iii) a conceptual dimension indexed by governance criteria and sub-criteria.

2.1.1 Panel Structure and Dimensionality

The dataset comprises observations of $N = 63$ Vietnamese provinces and centrally administered municipalities (hereafter, provinces) measured annually across $T = 14$ survey waves. The underlying constructs are measured through $K = 8$ dimension (Criteria hereafter, denoted C_k for $k = 1, \ldots, 8$), each composed of multiple sub-criteria (Sub-criteria hereafter, denoted SC_{jk} for $j = 1, \ldots, n_k$). In total, there are $p = 29$ sub-criteria across all criteria.

The data define a balanced panel in spatial ($N = 63$ provinces) and temporal ($T = 14$ years) dimensions, yielding a theoretical maximum of $M = N \times T = 882$ province-year

observations. Each observation indexed by (i, t) contains measurements of all $p = 29$ sub-criteria (subject to structural missingness, discussed in Section 2.2).

Data collection occurs annually during each survey wave. The survey is administered by Centre for Community Support & Development Studies (CECODES), the Vietnam Fatherland Front Research and Training Committee (VFF-CRT), and the United Nations Development Programme (UNDP) in collaboration with provincial statistical offices. Each annual wave typically surveys approximately 14,000 respondents across all 63 provinces, stratified by province, with final indices computed as province-level aggregates.

Survey data are collected and published on an annual calendar. The 14 waves span 2011–2024 inclusive:

$$\mathcal{T} = \{2011, 2012, \ldots, 2024\}$$

This period encompasses important shifts in Vietnam's governance environment, including the introduction of new indicators (environmental governance and e-governance added in 2018) and administrative disruptions resulting from provincial reorganizations and data collection challenges. The 14-year span provides sufficient temporal variation for forecasting model training.

The working format for weighting and ranking is a long-format panel with columns:

$$\text{panel_df} = \{\text{Province}, \text{Year}, \text{SC11}, \text{SC12}, \ldots, \text{SC83}\}$$

Thus, each row represents one province-year observation, with one row per 63 provinces per year, yielding up to 882 rows. Missing data and imputation are handled using strategies detailed in Section 2.2.

2.1.2 Measurement Framework and Indicator Definitions

All 29 sub-criteria are measured on a continuous scale with minimum 0 and practical maximum approximately 3.33 (reflecting the underlying survey instrument's design). Crucially, all sub-criteria are benefit-type indicators, meaning that higher values indicate stronger governance performance in the respective dimension. This uniform directionality has important implications for MCDM normalization and aggregation (discussed in Sections 2.3 and 2.4).

The hierarchical composition of criteria and sub-criteria is presented in Table 1. Each criterion encompasses 3–4 sub-criteria, comprehensively specifying the governance domains assessed:

Table 1: Comprehensive hierarchical structure of PAPI criteria and sub-criteria with measurement framework

Criteria	Criterion Name	Sub-Criterion	Sub-Criterion Name
C_1	Civic Participation	SC11	Civic Knowledge
		SC12	Opportunities for Participation in Elections
		SC13	Quality of Village Head Elections
		SC14	Voluntary Contributions
C_2	Transparency of Local Decision-Making	SC21	Access to Information
		SC22	Poverty List Transparency
		SC23	Communal Budget Transparency
		SC24	Transparent Land-Use Plan/Price Frames
C_3	Vertical Accountability	SC31	Interactions With Local Authorities
		SC32	Governments Responsive to Citizen Appeals
		SC33	Access to Justice Services
C_4	Control of Corruption in Public Sector	SC41	Limits on Corruption in Local Governments
		SC42	Limits on Corruption in Service Delivery
		SC43	Equity in State Employment
		SC44	Willingness to Fight Corruption
C_5	Public Administrative Procedures	SC51	Certification Procedures
		SC52	Construction Permits
		SC53	Land Title Procedures
		SC54	Personal Procedures
C_6	Public Service Delivery	SC61	Public Healthcare
		SC62	Primary Education
		SC63	Basic Infrastructure
		SC64	Law and Order
C_7	Environmental Governance	SC71	Environmental Protection
		SC72	Quality of Air
		SC73	Quality of Water
C_8	E-Governance	SC81	Access to E-Government Portals
		SC82	Access to the Internet
		SC83	E-Responsiveness of Provincial Authorities

The PAPI sub-criteria capture multiple dimensions of provincial governance capacity and citizen perception. Criterion C_1 (Civic Participation) reflects the extent and inclusiveness of civic engagement in provincial affairs, encompassing electoral participation and consultation mechanisms. Criterion C_2 (Transparency of Local Decision-Making) assesses the accessibility of provincial government information and transparency of budget execution and policy decisions. Criterion C_3 (Vertical Accountability) measures government responsiveness to citizen grievances and access to justice mechanisms for redress. Criterion C_4 (Control of Corruption in the Public Sector) captures the perceived effectiveness of anti-corruption

measures and equitable treatment in service delivery. Criterion C_5 (Public Administrative Procedures) evaluates the efficiency and simplification of administrative procedures and the burden of compliance on businesses and individuals. Criterion C_6 (Public Service Delivery) measures the quality and accessibility of province-delivered public services including healthcare, education, infrastructure, and utilities. Criterion C_7 (Environmental Governance) reflects environmental protection, sustainability of natural resources, and response to pollution and climate issues. Finally, Criterion C_8 (E-Governance) captures the availability of online government services, digital information systems, and internet accessibility.

2.2 Missing Data Structure and Data Imputation

The PAPI dataset exhibits substantial structural missingness reflecting both design constraints and administrative disruptions. The data comprise a balanced panel of 63 Vietnamese provinces measured across 14 annual survey waves (2011–2024) with 29 governance sub-criteria per province-year observation, yielding a theoretical maximum of 25,578 data cells ($63 \times 14 \times 29$). Across this domain, approximately 3,424 cells are missing, corresponding to a global missingness rate of 13.4%. While non-trivial, this missingness follows deterministic and partially predictable patterns that permit principled recovery. Understanding and properly addressing these missing data is essential to prevent bias propagation in the ensemble forecasting phase.

It is imperative to distinguish the analytical scope of the data imputation procedure within this study's bifurcated framework. The reconstructed dataset, generated through the MICE algorithm, is specifically and exclusively utilized to facilitate the ensemble machine learning forecasting phase, where maximum temporal and cross-sectional continuity is mathematically required for model training and stability. Conversely, the hierarchical CRITIC weighting and multi-criteria ranking procedures (Section 2.3 and 2.4) respect the empirical integrity of the initial dataset. These phases employ an adaptive exclusion mechanism that calculates composite indices based only on observed sub-criteria for each respective province-year observation, thereby ensuring that the resulting performance rankings are grounded in direct citizen experience rather than imputed values.

2.2.1 Missing Data Characterization and Quantification

Missing data in the PAPI panel arise from three distinct mechanisms, each with different statistical properties and implications for imputation strategy. Type 1 missing data reflect the structural evolution of the governance index itself, with new sub-criteria (environmental governance and e-governance indicators) formally introduced in 2018 and one sub-criterion (administrative procedures, SC52) discontinued from 2021 onward. Specifically, Environmental Governance sub-criteria (SC71, SC72, SC73) are absent for the entire 2011–2017 period, accounting for $3 \times 7 \times 63 = 1,323$ missing cells. E-Governance sub-criteria (SC81,

SC82, SC83) similarly were not measured from 2011–2017; SC83 became consistently available only from 2019 onward, contributing approximately 1,260 missing cells. The Transparency sub-criterion SC24 was absent in 2011–2017 and reintroduced in 2018 with partial coverage in certain provinces. Administrative Procedures (SC52) was entirely discontinued from 2021 onward, creating $4 \times 63 = 252$ missing cells in the final four years. Across these indicator-level absences, Type 1 missingness totals 3,116 cells, representing 90.9% of all missing data. These missings are deterministic and observable: the measurement agencies have explicit knowledge of which indicators had not been assembled in each year. This property makes Type 1 missingness conceptually Missing Completely At Random (MCAR) conditional on the observed year, since the absence is attributable to known governance index design choices rather than data collection failures or outcome characteristics.

Type 2 missingness comprises entire province-year observations where all 29 sub-criteria are absent. Nine such cases are identified, totaling 261 missing cells. In 2014, provinces P15 and P56 are entirely missing. Three provinces (P14, P15, P18) lack complete data in 2021. In 2023, provinces P14 and P47 are entirely absent; similarly, P17 and P52 are missing in 2024. These gaps reflect administrative disruptions in data collection, provincial-level survey failures, or data processing errors rather than structural index design. Under the assumption that such collection failures depend on observable administrative variables (provincial capacity, data collection infrastructure) rather on the governance quality being measured, Type 2 missingness is assessed as MCAR conditional on administrative covariates. This characterization supports the use of conditional imputation methods such as predictive models trained on provinces that did report in the given year, or within-province temporal interpolation.

Type 3 missingness comprises isolated missing cells within otherwise complete or substantially populated province-year observations. Four province-year combinations exhibit this pattern. In 2018, provinces P14 and P56 each are missing eight sub-criteria (SC21–SC24 and SC41–SC44), concentrated in the Transparency and Control of Corruption domains, suggesting category-specific non-response—possible survey administration failures affecting specific question blocks in particular provinces. In 2022, province P15 is missing 16 cells spanning Criteria 1, 2, 4, and 5, while province P18 is missing 15 cells concentrated in Criteria 2, 3, 4, and 8. Type 3 consists of only 47 cells (1.4% of total missingness) and likely reflects individual questionnaire item non-response or isolated data entry errors. If the propensity for item-level non-response depends on variables captured in the survey (respondent literacy, internet access, familiarity with governance topics), then Type 3 missingness is assessed as Missing At Random (MAR). Under this mechanism, MICE imputation including these auxiliary variables yields unbiased recovery (Rubin, 1987).

An overview of the annual missing data distribution across all three types is presented in Table 2. The inventory reveals that 2019 and 2020 are the only complete years with zero missing cells. The year 2018 marks the transition point where environmental and e-governance indicators entered the index, reducing single-indicator missings (Type 1) from

441 cell-years to 63. The bulk of Type 2 missingness occurs in 2014, 2021, 2023, and 2024, corresponding to known administrative events (reorganizations, survey disruptions) in certain provinces. Type 3 missingness, though small in absolute count, indicates localized data quality issues in specific province-year-criterion combinations.

Table 2: Missing data inventory by type and year

Year	Type 1	Type 2	Type 3	Total
2011	441	0	0	441
2012	441	0	0	441
2013	441	0	0	441
2014	427	58	0	485
2015	441	0	0	441
2016	441	0	0	441
2017	441	0	0	441
2018	63	0	16	79
2019	0	0	0	0
2020	0	0	0	0
2021	60	87	0	147
2022	61	0	31	92
2023	61	58	0	119
2024	61	58	0	119
Total	**3,116**	**261**	**47**	**3,424**

Type 3 missingness—scattered cells within otherwise substantially observed province-years—is limited to four province-year combinations: in 2018, provinces P14 and P56 each lack eight sub-criteria concentrated in Transparency and Corruption Control; in 2022, P15 lacks 16 cells spanning participation, transparency, corruption, and administrative procedures, while P18 lacks 15 cells in similar domains. These localised absences suggest category-specific non-response or data entry anomalies rather than complete survey failure.

2.2.2 Multivariate Imputation by Chained Equations

The missing data across all three types are recovered using Multivariate Imputation by Chained Equations (MICE), a principled statistical framework that estimates missing values by leveraging the multivariate interdependence structure among governance sub-criteria (Rubin, 1987; van Buuren, 2018). MICE supersedes simpler univariate methods—such as temporal forward-fill, cross-sectional mean imputation, or last-observation-carried-forward—by respecting the correlation structure among criteria and accommodating both point estimation and propagation of imputation uncertainty through multiple imputations (van Buuren, 2018). For a dataset like PAPI with moderate missingness ($\rho \approx 0.134$), well-defined feature relationships (governance sub-criteria are inherently correlated through latent governance quality), and mixed missingness mechanisms (Type 1 MCAR-by-design, Type 2/3 potentially MAR), MICE provides a theoretically justified and empirically robust approach.

MICE operates under explicitly stated assumptions regarding the missingness mechanism. The fundamental classification distinguishes three regimes. Missing Completely At Random (MCAR) holds when:

$$P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}, X_{\text{miss}}) = P(\text{Missing}(X_j) = 1) \tag{1}$$

that is, missingness depends on neither observed nor unobserved data. Under MCAR, complete-case analysis introduces no bias, only loss of statistical power. Missing At Random (MAR) permits:

$$P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}, X_{\text{miss}}) = P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}) \tag{2}$$

where missingness may depend on observed variables but not on the unobserved missing values themselves. For example, environmental governance data absent in 2011–2017 represents observed temporal information unrelated to the unobserved environmental quality in those years, a MAR scenario. Finally, Missing Not At Random (MNAR) violates MAR:

$$P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}, X_{\text{miss}}) \neq P(\text{Missing}(X_j) = 1 \mid X_{\text{obs}}) \tag{3}$$

occurring when missingness depends inherently on unobserved values—for instance, if corruption indicators were underreported when actual corruption is high. Under MNAR, standard imputation methods including MICE become biased without strong domain assumptions (van Buuren, 2018).

For the PAPI dataset, the three identified missing data types are evaluated as follows. Type 1 (Structural Index Expansion) stems from known governance index design choices: Environmental and E-Governance indicators absent in 2011–2017 by foundational design, not by data inadequacy. This is deterministic and declaratively MCAR conditional on the observed year. Type 2 (Entire Province-Year Gaps) reflects administrative disruptions—provincial data collection failures, survey non-response—plausibly attributable to observable administrative or logistical factors (provincial capacity, data collection infrastructure state) rather than the governance construct being measured. These cases are conservatively treated as MCAR conditional on administrative covariates. Type 3 (Scattered Cells) consists of individual questionnaire item non-responses, assessed as MAR if the propensity for non-response depends on respondent-level characteristics (literacy, internet familiarity, governance engagement) recorded in survey auxiliary variables, all of which will be included in the MICE imputation model. Under combined MCAR/MAR assumptions—which hold plausibly for PAPI —MICE yields unbiased or near-unbiased recovery of missing values (van Buuren, 2018).

The MICE algorithm operates on a data matrix $X = (x_1, x_2, \ldots, x_p) \in \mathbb{R}^{n \times p}$ containing n observations and p features with missing values. Initialization fills all missing entries with

column-wise empirical means:

$$X_{ij}^{(0)} = \begin{cases} X_{ij} & \text{if } X_{ij} \text{ observed} \\ \bar{x}_j = \frac{1}{n}\sum_{i'=1}^{n} X_{i'j} & \text{if } X_{ij} \text{ missing} \end{cases} \tag{4}$$

This initialization provides starting values for the iterative imputation loop. The algorithm then cycles through iterations $t = 1, 2, \ldots, T_{\text{max}}$ (default: $T_{\text{max}} = 20$). At each iteration and for each feature $j = 1, \ldots, p$: (1) the missing entries of feature x_j are temporarily restored to missing status; (2) a regression model is fitted on the complete cases using all other features as predictors and the observed values of x_j as the target:

$$\hat{f}_j^{(t)} = \text{Regressor}(X_{\text{obs},-j}^{(t-1)}, X_{\text{obs},j}^{(t-1)}) \tag{5}$$

(3) the fitted model predicts imputed values for the missing entries:

$$X_{\text{miss},j}^{(t)} = \hat{f}_j^{(t)}(X_{\text{miss},-j}^{(t-1)}) \tag{6}$$

and (4) feature j is updated with both observed and imputed values for use in subsequent feature iterations:

$$X_{\cdot j}^{(t)} = \begin{cases} X_{\text{obs},j}^{(t-1)} & \text{observed cells} \\ X_{\text{miss},j}^{(t)} & \text{imputed cells} \end{cases} \tag{7}$$

For the regression component, this study employs ExtraTreesRegressor (Extremely Randomized Trees) following guidance in the imputation literature (Doove et al., 2014). ExtraTreesRegressor is selected because it captures nonlinear relationships among governance criteria without assuming parametric form; exhibits robustness to scale heterogeneity since governance metrics span different measurement scales and distributions; achieves rapid convergence with tree-based ensembles typically stabilizing by iterations 10–15; and integrates seamlessly with scikit-learn's `IterativeImputer` implementation (Pedregosa et al., 2011). A central theoretical result establishes monotone improvement in the likelihood of observed data across iterations. Under the assumption that the posterior distribution $p(X_{\text{miss}} \mid X_{\text{obs}})$ admits consistent estimation via the regression models employed, the log-likelihood of observed data satisfies (van Buuren, 2018; van Buuren & Groothuis-Oudshoorn, 2011):

$$\ell(X_{\text{obs}} \mid \hat{\theta}^{(t+1)}) \geq \ell(X_{\text{obs}} \mid \hat{\theta}^{(t)}) \quad \text{for all } t \geq 1 \tag{8}$$

This monotonicity theorem guarantees that each iteration improves (or maintains) the fit to observed data, ensuring that imputed values become progressively more plausible. Convergence rate depends on missingness proportion (the PAPI rate of 13.4% facilitates rapid convergence), feature correlation structure (highly correlated governance criteria enable rapid imputation), and estimator choice (tree-based methods typically converge by iteration 15,

parametric models may require 50+ iterations).

Modern imputation practice employs multiple imputation ($M > 1$ complete datasets) rather than single imputation to quantify imputation uncertainty (Rubin, 1987). The procedure involves: (1) running the MICE algorithm M times with different random seeds to generate M independent complete datasets $\{X^{(1)}, X^{(2)}, \ldots, X^{(M)}\}$; (2) analyzing each imputed dataset independently to obtain M point estimates $\{\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(M)}\}$; and (3) pooling estimates via Rubin's Rules (Rubin, 1987). The overall pooled point estimate is the arithmetic mean of the M analysis-specific estimates:

$$\hat{\theta}_{\text{pooled}} = \frac{1}{M} \sum_{m=1}^{M} \hat{\theta}^{(m)} \tag{9}$$

The total variance reflects both within-imputation variance (averaged across the M datasets) and between-imputation variance (variability of estimates across imputations):

$$\bar{V}_{\text{within}} = \frac{1}{M} \sum_{m=1}^{M} \hat{V}^{(m)} \quad ; \quad V_{\text{between}} = \frac{1}{M-1} \sum_{m=1}^{M} \left(\hat{\theta}^{(m)} - \hat{\theta}_{\text{pooled}} \right)^2 \tag{10}$$

$$\hat{V}_{\text{total}} = \bar{V}_{\text{within}} + \left(1 + \frac{1}{M} \right) V_{\text{between}} \tag{11}$$

For the PAPI weighting and ranking analyses, we employ $M = 5$ imputations, the standard in applied multiple imputation practice (van Buuren, 2018), producing point estimates along with principled uncertainty quantification.

Implementation details are summarized in Table 3. All imputation is executed via scikit-learn's `sklearn.impute.IterativeImputer` with ExtraTreesRegressor as the estimator, $T_{\text{max}} = 20$ iterations, 100 trees, maximum tree depth of 6 (to prevent overfitting to imputation residuals), minimum leaf size of 3 (to ensure predictions reflect diverse training examples), and a fixed random seed of 42 for reproducibility. Beyond imputation, binary missingness indicators $\text{WasMissing}_j \in \{0, 1\}$ are appended as additional features for each originally missing column, signalling which cells were imputed. These indicators are incorporated into downstream forecasting models, allowing predictive algorithms to learn and account for differential uncertainty between imputed and observed values, an approach validated by empirical evidence showing model improvements when explicit imputation signals are provided (van Buuren, 2018).

2.2.3 Post-Imputation Data Quality

After MICE imputation, the dataset contains no missing values. To validate imputation quality, we conduct sensitivity analyses examining robustness to alternative imputation schemes. Preliminary analyses confirm that:

Table 3: MICE implementation hyperparameters

Parameter	Value	Rationale
Estimator	ExtraTreesRegressor	Nonlinear, robust, fast
Max iterations	20	Achieves convergence for $\rho \approx 0.13$
n-estimators (trees)	100	Sufficient for stable predictions
Max depth	6	Prevents overfitting to imputation residuals
Min samples leaf	3	Ensures diverse predictions
Random state	42	Reproducibility

- Sub-criterion means shift $< 2\%$ when using alternative MICE estimators (RandomForest, BayesianRidge).
- Criterion-level weights (Section 2.3) shift $< 5\%$ under MNAR sensitivity scenarios with plausible misspecification ($\delta \in [-0.5, +0.5]$).
- Ranking correlation with single-imputation (M = 1) exceeds 0.98, indicating negligible ranking reordering from multiple imputation variance.

These findings support the robustness and stability of results to imputation methodological choices. Summary statistics for sub-criteria across all provinces and years (Table 4) are reported after imputation. Before imputation, structural missingness (13.4% of all cells; see earlier discussion) distorts descriptive statistics; post-imputation statistics reflect distributional properties after handling.

Table 4: Post-imputation descriptive statistics of PAPI criteria)

Criterion	Mean	Std. Dev.	Min	Max
C_1 (Participation)	2.41	0.34	1.12	3.28
C_2 (Transparency)	2.18	0.38	0.89	3.10
C_3 (Accountability)	2.35	0.29	1.45	3.15
C_4 (Anti-Corruption)	2.26	0.32	1.08	3.02
C_5 (Admin Procedures)	2.54	0.36	1.20	3.25
C_6 (Service Delivery)	2.48	0.31	1.35	3.18
C_7 (Environmental)	2.33	0.41	0.65	3.16
C_8 (E-Governance)	2.12	0.37	0.52	3.05

The adoption of a multivariate recursive partitioning approach for missing data recovery ensures that the PAPI panel is transformed into a rigorous, balanced longitudinal structure suitable for high-precision performance assessment. By preserving the inherent cross-sectional and temporal correlation structures through MICE with extremely randomized trees, the imputation strategy mitigates the risk of estimation bias and restores the statistical power of the dataset for the ensemble machine learning forecasting phase. As previously established, while the weighting and ranking analyses operate on the original dataset through

adaptive missing-value handling, the forecasting phase leverages this fully populated evidentiary base to ensure predictive reliability and mathematical stability.

2.3 CRiteria Importance Through Inter-criteria Correlation

The Criteria Importance Through Intercriteria Correlation (CRITIC) method (Diakoulaki et al., 1995) is an objective weighting procedure that derives criterion importances directly from data heterogeneity and interdependence, without subjective expert elicitation. This study extends CRITIC from its traditional single-level application to a novel two-level hierarchical framework tailored to the multi-tier structure of the PAPI dataset.

Traditional weighting schemes in governance indices rely on equal weighting: each sub-criterion receives $1/n_k$ within its criterion, and each criterion receives $1/K$ globally. While transparent and easily communicated, equal weighting ignores two critical sources of information in the data: (i) **contrast intensity**: a sub-criterion with high variance across provinces carries more discriminatory power than one where all provinces score similarly; and (ii) **redundancy**: two sub-criteria with high correlation carry overlapping information and should not receive full joint weight.

The CRITIC method operationalizes these principles by computing an "informativeness score" for each criterion that rewards both high variance and low correlation with peers, then normalizing these scores into weights. This provides a principled alternative to equal weighting that respects the effective dimensionality of the dataset.

2.3.1 Two-Level CRITIC: Hierarchical Weight Derivation

Foundational Framework. The CRITIC method addresses a fundamental problem in multi-dimensional governance assessment: identifying which dimensions carry unique discriminatory information. For a decision matrix $X \in \mathbb{R}^{m \times n}$ with m observations and n criteria/sub-criteria, the CRITIC informativeness score for criterion j is:

$$C_j = \sigma_j \sum_{j'=1}^{n} (1 - r_{jj'}) \tag{12}$$

This score simultaneously rewards two desirable properties. First, $\sigma_j = \text{std}(X_{\cdot j})$ captures *contrast intensity*—the degree to which criterion j discriminates across provinces. A criterion where all provinces achieve identical scores ($\sigma_j = 0$) provides no ranking information; such criteria receive zero weight. Conversely, high-variance criteria effectively partition provinces into hierarchically distinct groups. Second, the term $\sum_{j'=1}^{n}(1 - r_{jj'})$ measures *informational uniqueness* by summing the non-correlation of criterion j with all peers. This term quantifies the degree to which criterion j captures information not already captured by other criteria. For $j' = j$, $r_{jj} = 1$ by definition, yielding $(1 - 1) = 0$. For $j' \neq j$: when two criteria are highly positively correlated ($r_{jj'} \to 1$), the term $(1 - r_{jj'}) \to 0$, signaling that

criterion j carries largely redundant information; conversely, when criteria are uncorrelated or negatively correlated, the term is maximized, indicating unique information content.

The normalized weight for criterion j is then:

$$w_j = \frac{C_j}{\sum_{j'=1}^{n} C_{j'}} \tag{13}$$

These weights form a probability simplex: $\sum_{j=1}^{n} w_j = 1$ and $w_j \geq 0$ for all j, ensuring that weights are formally valid probability distributions suitable for aggregation.

Intuition and Information Theory Connection: The CRITIC framework can be interpreted through the lens of information theory. A criterion that is (1) highly variable carries high *Shannon entropy*, indicating capacity to differentiate observations; simultaneously, uncorrelated criteria contribute to *mutual information reduction* across the criterion set, meaning each criterion is less redundant. Criteria receiving near-zero weight are typically either (i) constant across all observations (no discriminatory power), (ii) highly redundant with peers (information already captured elsewhere), or (iii) both. Conversely, high-weight criteria are those that discriminate most effectively while conveying information not present elsewhere in the dataset.

Level 1: Local Sub-Criterion Weights per Criterion. A critical innovation in this framework is the two-stage decomposition, which solves a fundamental problem in hierarchical data: without hierarchy-aware weighting, CRITIC applied naively to all 29 sub-criteria simultaneously would conflate within-criterion importance (how central a sub-criterion is to its parent criterion's definition) with between-criterion importance (how much that criterion matters globally). This conflation can lead to over-weighting sub-criteria in weak-signal criteria or under-weighting sub-criteria in high-signal criteria. To avoid this, Level 1 applies CRITIC independently within each of the $K = 8$ criterion groups, treating each criterion's sub-criteria as a localized eigensystem.

For criterion group C_k:

1. **Extract sub-matrix:** Retain only the columns corresponding to sub-criteria belonging to C_k. Let $X_k \in \mathbb{R}^{m \cdot T \times n_k}$ denote this matrix of $m \times T$ observations (all provinces and years) and n_k sub-criteria (e.g., $n_1 = 4$ for Civic Participation). In practice, $m \times T \approx 756$ (63 provinces $\times$ 14 years after removing entirely missing rows). This extraction ensures that correlations computed in step 3 reflect only within-criterion redundancy, not cross-criterion correlations that belong to Level 2 analysis.

2. **Normalize to common scale:** Apply min-max normalization to place all sub-criteria on a common $[0, 1]$ scale:

$$\tilde{X}_{k,ij} = \frac{X_{k,ij} - \min_i X_{k,ij}}{\max_i X_{k,ij} - \min_i X_{k,ij} + \varepsilon} \tag{14}$$

where $\varepsilon = 10^{-10}$ is a small constant to prevent division by zero (or numerical instability) for near-constant columns. **Why normalize?** Sub-criteria within a criterion may be mea-

sured on different scales (e.g., Civic Knowledge on a 0–3.33 scale, Electoral Participation as a percentage). The standard deviation σ used in CRITIC (Eq. (15)) is scale-dependent; normalization ensures that all sub-criteria are comparable on a common dimension. Without normalization, a sub-criterion with naturally larger scale variance would artificially dominate, regardless of its actual discriminatory power relative to peers.

3. **Compute local CRITIC scores:** Calculate informativeness for each sub-criterion within C_k:

$$C_{k,j} = \sigma(\tilde{X}_{k,\cdot j}) \sum_{j'=1}^{n_k} \Big(1 - r(\tilde{X}_{k,\cdot j}, \tilde{X}_{k,\cdot j'})\Big) \tag{15}$$

This score reflects the combined discriminatory power and uniqueness of sub-criterion j within criterion k's local universe of n_k sub-criteria.

4. **Normalize to local weights:** Derive locally-normalized weights within the criterion group:

$$u_{k,j} = \frac{C_{k,j}}{\sum_{j'=1}^{n_k} C_{k,j'}} \quad \text{where} \quad \sum_{j=1}^{n_k} u_{k,j} = 1 \tag{16}$$

These weights, $\mathbf{u}_k = (u_{k,1}, \ldots, u_{k,n_k})^\top$, represent the *relative* importance of each sub-criterion within its parent criterion, forming a probability distribution over the n_k sub-criteria in group k. For any future projection onto this criterion, a weighted score is:

$$\text{Score}_{C_k} = \sum_{j=1}^{n_k} u_{k,j} \cdot \text{SC}_{kj} \tag{17}$$

Notably, $u_{k,j}$ answers the question: *"Within criterion C_k, what fraction of this criterion's 'information budget' should go to sub-criterion j?"*

Building the Criterion-Level Composite Matrix (Dimensionality Reduction with Hierarchy-Awareness). After computing Level 1 weights for all $K = 8$ criteria, a "bridge" matrix is constructed that represents each criterion as a single composite score, incorporating the within-group sub-criterion weights. For each province-year observation (i, t), the following is computed:

$$Z_{i,k} = \sum_{j=1}^{n_k} u_{k,j} \cdot X_{i,k,j} \tag{18}$$

where $u_{k,j}$ are the Level 1 local weights (from Equation (16)), and $X_{i,k,j}$ are the *original* (pre-normalized) sub-criterion values for province i in criterion k, sub-criterion j.

This yields a criterion-level matrix $Z \in \mathbb{R}^{m \cdot T \times K}$ with $K = 8$ columns (one per criterion) and approximately 756 rows (all usable province-year observations). In cases where one or more sub-criteria are missing for a specific province-year (i, t), the composite score $Z_{i,k}$ is computed by adaptively re-normalizing the local weights (*i.e.,* $u_{k,j} / \sum_{j \in \text{Obs}} u_{k,j}$ where Obs is

the set of observed indicators) to preserve the proportionality of the available measurements. **Critical design choice: Using original values, not normalized values.** Equation (18) deliberately uses the original $X_{i,k,j}$ values rather than the normalized $\tilde{X}_{k,ij}$ values. Why? If forced to use normalized values, the composite scores $Z_{i,k}$ would be artificially constrained to a narrow band (approximately $[0, 1]$), losing information about the true magnitude differences between criteria. Original values preserve the actual scale and distribution of each criterion: if Cultural Heritage scores range 0–3.5 and Economic Justice ranges 1.2–3.1, these distributional differences persist in Z, enabling Level 2 to accurately assess how criterion variances differ. This design prevents *double normalization*, which would compress information and distort Level 2's correlation structure.

Rationale and Separation of Information Layers: The composite matrix Z performs dimensionality reduction (from 29 dimensions to 8), guided by data-driven Level 1 weights rather than ad hoc averaging. This ensures two properties: (i) within-criterion redundancy is accounted for (high internal correlation → lower composite weight), and (ii) all 29 sub-criteria contribute to the final governance assessment (no criterion is entirely discarded). Level 2 then operates on this intelligently reduced representation, viewing inter-criterion information rather than raw sub-criterion information. This recursive hierarchy naturally separates concerns: Level 1 asks "Which sub-criteria within each criterion are most informative?" and Level 2 asks "Across criteria, which convey unique information about governance differentiation?"

Formally, the composite matrix Z respects the hierarchical information structure: variance in Z comes from (i) true differences in criterion scores and (ii) covariation between sub-criteria within each criterion (weighted by $u_{k,j}$). Correlation in Z reflects genuine inter-criterion dependence unconfounded by within-criterion sub-criterion multicollinearity.

Level 2: Global Criterion Weights (Inter-Criterion Information Assessment). CRITIC is now applied a second time to the criterion-level composite matrix Z, shifting focus from within-criterion to between-criterion relationships:

1. **Extract and prepare matrix:** Take $Z \in \mathbb{R}^{m \cdot T \times 8}$ with 8 criterion columns and 756 rows.

2. **Normalize to common scale:** Apply min-max normalization to each criterion column:

$$\tilde{Z}_{i,k} = \frac{Z_{i,k} - \min_i Z_{i,k}}{\max_i Z_{i,k} - \min_i Z_{i,k} + \varepsilon} \tag{19}$$

Again, normalization ensures comparability: criterion composite scores may naturally span different ranges, and normalization ensures that CRITIC treats each on an equal footing.

3. **Compute global CRITIC scores:** Calculate informativeness for each of the 8 criteria:

$$C_k = \sigma(\tilde{Z}_{\cdot k}) \sum_{k'=1}^{K} \Big(1 - r(\tilde{Z}_{\cdot k}, \tilde{Z}_{\cdot k'})\Big) \tag{20}$$

This score captures: (i) *variance of criterion* k: $\sigma(\tilde{Z}_{\cdot k})$ reflects how much criterion k's composite score differentiates provinces (after within-criterion redundancy has been removed), and (ii) *inter-criterion uniqueness*: $\sum_{k'=1}^{K}(1 - r(\tilde{Z}_{\cdot k}, \tilde{Z}_{\cdot k'}))$ measures how much criterion k provides information not already captured by other criteria. Critically, at this stage, correlation is computed across criteria (not sub-criteria), so a high $1 - r_{kk'}$ value means "criterion k varies independently of criterion k'," indicating minimal redundancy at the criterion level.

4. **Compute global criterion weights:**

$$v_k = \frac{C_k}{\sum_{k'=1}^{K} C_{k'}} \quad \text{where} \quad \sum_{k=1}^{K} v_k = 1 \tag{21}$$

These weights, $\mathbf{v} = (v_1, \ldots, v_K)^\top$, represent the *global importance coefficient* for each criterion, reflecting its discriminatory power and informational uniqueness at the criterion level. They directly answer: *"After accounting for within-criterion redundancy (Level 1), which criteria are most informative about inter-provincial governance differentiation?"* Criteria with high v_k are those whose composite scores (already optimized by Level 1 weighting) exhibit both large variance and low correlation with other criteria. Criteria with low v_k either have low discriminatory power (most provinces achieve similar composite scores) or are highly redundant with peer criteria (composite score information is already captured elsewhere).

Hierarchical Weight Synthesis via Product Rule. The final set of 29 global sub-criterion weights is derived by combining Level 1 and Level 2 information via the product rule:

$$w_j = u_{k,j} \cdot v_k \quad \text{for all sub-criteria } j \in \text{SC}_k \tag{22}$$

This product decomposes global importance into multiplicative components: each global weight w_j is the product of (i) its local rank within its parent criterion ($u_{k,j}$, the "within-criterion importance") and (ii) the importance of that parent criterion globally (v_k, the "global criterion importance"). **Hierarchical meaning:** This decomposition ensures non-compensatory hierarchical constraints:

- *No within-level compensation:* A sub-criterion with the highest $u_{k,j}$ in a low-v_k criterion cannot achieve high global weight, because multiplying a large local weight by a small global criterion weight yields a small global sub-criterion weight. Conversely, even a modest local weight inherits the importance of a globally important criterion.
- *Interpretability:* The product form is intuitive: a physician's sub-criterion (e.g., "quality of maternal healthcare") under Criterion 6 (Public Service Delivery) inherits importance both from (a) being the highest-variance healthcare indicator (high $u_{k,j}$) and (b) healthcare delivery being a critical determinant of provincial governance (high v_k). If either factor is weak, the global weight is modest; both must be strong for the global weight to be prominent.

- *Information preservation:* All 29 sub-criteria retain positive weights (since $u_{k,j} > 0$ and $v_k > 0$ for all k, j), ensuring that every dimension of the governance index contributes to the final assessment, albeit with data-driven proportionality.

Formal Constraint Satisfaction (Simplex Preservation Theorem): The global weights automatically satisfy the fundamental probability simplex constraint:

$$\sum_{k=1}^{K}\sum_{j=1}^{n_k} w_j = \sum_{k=1}^{K} v_k \underbrace{\sum_{j=1}^{n_k} u_{k,j}}_{=1} = \sum_{k=1}^{K} v_k = 1 \tag{23}$$

Proof sketch: For each criterion k, by definition Level 1 weights satisfy $\sum_{j=1}^{n_k} u_{k,j} = 1$ (they are normalized on criterion k's simplex). Summing the global weights across all sub-criteria in criterion k yields $\sum_{j=1}^{n_k} w_j = v_k \cdot 1 = v_k$. Summing across all criteria $k = 1, \ldots, K$ then yields $\sum_{k=1}^{K} v_k = 1$, since v_k are Level 2 weights normalized to the global simplex. Thus, global weights sum exactly to unity without ad hoc re-normalization (though a floating-point numerical guard is typically applied for computational stability), confirming that the hierarchical product rule preserves the necessary probabilistic structure.

Final Aggregation and Analysis Pipeline. These 29 globally-normalized weights are now passed to both the Hierarchical MCDM ranking pipeline (Section 2.4) and to the ensemble forecasting pipeline (Section 2.5). The two-level framework, represented by weights $\mathbf{w} = (w_1, \ldots, w_{29})^\top$, constitutes the objective, data-driven prioritization of information across all governance dimensions.

2.3.2 Weighting Analysis

Temporal Stability Analysis: Window-Based Assessment

The split-half test provides a binary temporal stability verdict. To capture more nuanced temporal dynamics and characterize weight evolution across the full 14-year window, we employ a window-based approach that extracts multiple overlapping time windows and computes agreement statistics across them. This approach is more statistically powerful than binary tests and provides actionable information about *when* and *where* weight shifts occur.

Motivation: A core assumption in governance assessment is that the prioritization of criteria—their relative importance for discriminating provinces—remains stable across time. Substantial drift in weights over 14 years would suggest either cumulative data quality degradation, genuine structural breaks in governance patterns requiring time-variant analysis, or both. A robust weighting system should exhibit temporal consistency, enabling the use of a single unified weight vector across all observation years.

Methodological approach: Rather than a single binary temporal stability test, we employ a window-based approach that extracts multiple overlapping time windows and computes rank agreement metrics across them. This approach is statistically more powerful than

binary tests and provides disaggregated information identifying *when* and *where* weight instability emerges, should it occur.

Window construction: The 14-year panel (2011–2024) is decomposed into overlapping time windows of fixed length $L = 5$ years with unit step size (consecutive windows overlap by 4 years). This yields $(T - L) + 1 = 10$ distinct windows:

$$\text{Window}_k = [\text{year}_{\text{min}} + (k-1), \ldots, \text{year}_{\text{min}} + (k-1) + 4] \quad \text{for } k = 1, 2, \ldots, 10 \tag{24}$$

For each window, a mean weight vector is computed by averaging the criterion weights across the constituent years:

$$\bar{\mathbf{w}}_k = \frac{1}{L} \sum_{t \in \text{Window}_k} \mathbf{w}_t \tag{25}$$

This produces 10 window-specific mean weight vectors, each representing the average prioritization scheme in a 5-year epoch. Comparisons among these vectors reveal medium-term weight trajectories and any persistent deviations.

Spearman's rank correlation across windows: The ordinal ranking of criteria may drift due to temporal shifts in governance dynamics. To quantify this drift sensitively, we compute Spearman's rank correlation coefficient ρ between consecutive windows. For windows k and $k+1$, define $\text{rank}_k(j)$ as the ordinal rank of criterion j when all 29 criteria are sorted by mean weight $\bar{w}_{k,j}$ (ascending). Then:

$$\rho_k = 1 - \frac{6 \sum_{j=1}^{n} (\text{rank}_k(j) - \text{rank}_{k+1}(j))^2}{n(n^2 - 1)} \tag{26}$$

This yields $W - 1 = 9$ pairwise rank correlations across consecutive windows. High mean correlation (e.g., $\bar{\rho} > 0.75$) signals stable criterion orderings; deviations from this threshold indicate reorderings worthy of investigation. The standard deviation σ_ρ of correlations across windows captures volatility, identifying whether instability is episodic or persistent.

Kendall's W: Omnibus agreement coefficient: While pairwise Spearman correlations characterize adjacent-window agreement, we employ Kendall's coefficient of concordance W to obtain a single omnibus statistic measuring agreement across *all* 10 windows simultaneously. This statistic quantifies the global consistency of criterion rankings across the full temporal span:

$$W = \frac{12S}{m^2(n^3 - n)} \tag{27}$$

where $m = 10$ (number of windows), $n = 29$ (number of criteria), and $S = \sum_{j=1}^{n} (R_j - \bar{R})^2$ is the sum of squared rank deviations, with $\bar{R} = \frac{m(n+1)}{2}$ as the theoretical mean rank. The coefficient ranges $W \in [0, 1]$: $W = 1$ indicates identical criterion rankings in all windows

(perfect agreement), while $W = 0$ indicates independent, random rankings. For the PAPI dataset, W serves as a global temporal stability indicator.

Coefficient of variation analysis: While Spearman's ρ and Kendall's W measure ranking position stability, they are insensitive to changes in weight magnitudes. The Coefficient of Variation (CV) addresses this by capturing absolute weight variability. For each criterion j, compute:

$$\text{CV}_j = \frac{\sigma_j}{\bar{w}_j} \tag{28}$$

where σ_j and $\bar{w}_j$ are the standard deviation and mean of criterion j's weight across all 14 individual years. CV values range $[0, \infty)$: CV < 0.10 indicates highly stable weight magnitude; CV $\in [0.10, 0.30]$ indicates moderate variation; CV > 0.30 signals substantial weight volatility. Computing CV for all 29 criteria yields a per-criterion sensitivity profile identifying which criteria exhibit most/least temporal stability. Criteria with low CV are reliable components of the weighting scheme, while high-CV criteria warrant additional scrutiny.

Interpretation and reporting: The window-based temporal stability analysis produces three complementary outputs: (i) a time-series of 9 pairwise Spearman correlations, enabling identification of specific periods where ranking instability emerges; (ii) a single Kendall's W statistic summarizing global agreement across all windows; and (iii) a per-criterion CV profile revealing which criteria are most temporally fragile. These metrics collectively establish whether the weight vector exhibits sufficient temporal consistency to justify its use as a unified parameter across all years 2011–2024, or whether time-variant weighting would be warranted.

Sensitivity Analysis: Three-Tier Perturbation Framework

Motivation and complementarity: While temporal stability assesses parameter consistency over time, sensitivity analysis addresses an orthogonal robustness question: *How resilient are the final rankings to perturbations in weight values?* This resilience is essential because weight uncertainty arises from multiple sources: (i) measurement error in the underlying governance sub-criteria, (ii) model specification ambiguity in the hierarchical CRITIC procedure, (iii) legitimate alternative governance prioritizations reflecting different stakeholder preferences, and (iv) sampling variability if the analysis were repeated with different cohorts or years. Sensitivity analysis quantifies how much ranking positions would change under plausible deviations from the point-estimate CRITIC weights.

Three-tier framework design: We employ a Monte Carlo perturbation framework with three progressively intense perturbation magnitudes, progressively stressing the weight vector:

- **Conservative tier ($\pm$5%):** Reflects small measurement errors or minor model variations. If rankings prove stable under $\pm$5% weight perturbations, the weighting scheme is robust to small specification uncertainties.

- **Moderate tier ($\pm$15%):** Reflects realistic policy revisions or substantial criterion data revisions (e.g., 2018 governance index redesign). Stability under $\pm$15% perturbations suggests resilience to non-trivial weight changes.
- **Aggressive tier ($\pm$50%):** Represents extreme stress testing. Massive weight deviations ($\pm$50%) simulate worst-case scenarios where governance assessment paradigms shift radically. Robustness here is unexpected but highly desirable.

Perturbation procedure: For each tier $t \in \{\text{conservative}, \text{moderate}, \text{aggressive}\}$, conduct $R = 1000$ independent Monte Carlo replicates. For replicate r:

1. Generate random perturbations $\delta_{r,j} \sim \text{Uniform}(-m_t, +m_t)$ for each criterion j, with $m_t \in \{0.05, 0.15, 0.50\}$ corresponding to the three tiers.
2. Apply perturbations to compute perturbed weights: $\tilde{w}_{r,j} = w_j^*(1 + \delta_{r,j})$.
3. Apply non-negativity guard to ensure computational stability: $\tilde{w}_{r,j} := \max(\tilde{w}_{r,j}, 10^{-8})$.
4. Re-normalize to enforce the probability simplex constraint $\sum_j \hat{w}_{r,j} = 1.0$:

$$\hat{w}_{r,j} = \frac{\tilde{w}_{r,j}}{\sum_{\ell=1}^{n} \tilde{w}_{r,\ell}} \tag{29}$$

5. Re-rank all criteria by perturbed weights and compute the rank ordering change relative to the original.

Rank disruption metric: For each replicate, quantify the magnitude of ranking disruption using Spearman's rank correlation between the original and perturbed criterion orderings:

$$\text{Disruption}_r = 1 - \rho(\text{rank}_{\text{original}}, \text{rank}_{\text{perturbed},r}) \tag{30}$$

The disruption metric ranges $[0, 1]$: a value near 0 indicates that ranking remains stable despite perturbation (robust to weight variations), while values near 1 indicate substantial rank reordering (sensitive to weight specification). Aggregating across all 1000 replicates per tier yields the mean disruption $\overline{\text{Disruption}}_t$ for each tier t.

Robustness scoring: Define the robustness score for perturbation tier t as:

$$\text{Robustness}_t = 1 - \overline{\text{Disruption}}_t \in [0, 1] \tag{31}$$

By construction, robustness should decrease monotonically as perturbation intensity increases:

$$\text{Robustness}_{\text{conservative}} \geq \text{Robustness}_{\text{moderate}} \geq \text{Robustness}_{\text{aggressive}}$$

This monotonicity property is both expected and diagnostic: violation of the monotonicity order signals either numerical instability or weight configurations prone to tie-breaking

sensitivity, warranting investigation. For a well-designed weighting system, conservative tier robustness should remain high (>0.85), indicating resilience to minor perturbations.

Per-criterion sensitivity profiles: Aggregation at the tier level masks criterion-specific patterns. To identify "fragile" criteria prone to rank reordering, compute for each criterion j and tier t the mean rank position change across all replicates:

$$\Delta\text{rank}_{t,j} = \frac{1}{R}\sum_{r=1}^{R} |\text{rank}_{\text{original}}(j) - \text{rank}_{\text{perturbed}}(r,j)| \tag{32}$$

Values near 0 indicate ranking stability; larger values (up to $n-1=28$) indicate frequent rank shifts. Criteria with high Δrank are *sensitive* (their rank position fluctuates readily with weight changes), while those with low Δrank are *robust* (their ranking remains anchored). Computing this profile across all 29 criteria and 3 tiers produces a detailed sensitivity matrix revealing which criteria are most vulnerable to weight specification uncertainty. Such criteria merit careful review and may require additional validation before deployment in critical governance assessments.

2.4 MCDM Ranking Methods

To construct a robust performance index that is invariant to methodological idiosyncrasies, we employ a multi-methodological framework comprising five distinct Multi-Criteria Decision Making (MCDM) algorithms. A known limitation of composite indexing is the sensitivity of the final ranking to the aggregation technique (Greco et al., 2016). By generating independent rankings utilizing distance-based, outranking, and proportional assessment paradigms, we establish a robust consensus evaluation that mitigates the risk of any single algorithm's mathematical artefacts dominating the governance ranking.

Let $\tilde{X} \in \mathbb{R}^{m\times n}$ represent the performance matrix, where $\tilde{x}_{ij} \in [0,1]$ denotes the min-max normalized score of province i on sub-criterion j (as defined in Section 2.3), and w_j constitutes the corresponding global CRITIC weight such that $\sum_j w_j = 1$. Consistent with the weighting phase, the MCDM ranking framework utilizes the original dataset, excluding missing values from the normalization and distance-based calculations. For province-year observations with missing data (primarily for environmental and e-governance indicators prior to 2018), the algorithms adaptively re-scale the global weights $\mathbf{w}$ to ensure that the composite index represents a valid weighted average of the observed performance dimensions. This methodology prevents the propagation of imputation uncertainty into the normative ranking of provincial governance performance. Given that the PAPI assessment framework operates strictly with benefit-type indicators (as detailed in Table 1), cost-type formulations in the following algorithms are presented for generalized theoretical completeness but are effectively constrained to the benefit subspace in empirical application.

2.4.1 TOPSIS

The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) establishes rankings based on the geometric distance of an alternative from hypothetical ideal states (Hwang & Yoon, 1981). The foundational premise of TOPSIS is that the optimal alternative should simultaneously minimize the distance to the Positive Ideal Solution (PIS) and maximize the distance from the Negative Ideal Solution (NIS). This simultaneous geometric optimization is highly effective in differentiating alternatives that might otherwise appear similar under simple linear aggregation.

First, the normalized performance matrix is mapped into a weighted metric space: $v_{ij} = w_j \tilde{x}_{ij}$. The PIS ($A^+$) and NIS ($A^-$) vectors are determined by extracting the extremum performance values across all observed provinces:

$$A^+ = \{v_1^+, v_2^+, \ldots, v_n^+\} = \left\{\max_i v_{ij}\right\} \tag{33}$$

$$A^- = \{v_1^-, v_2^-, \ldots, v_n^-\} = \left\{\min_i v_{ij}\right\} \tag{34}$$

We measure the separation of each province i from these ideal solutions using the Euclidean distance metric (L_2 norm). The L_2 norm implies a compensatory relationship where severe underperformance in one criterion can be partially offset by strong outperformance in another, though the quadratic penalty disproportionately heavily weights extreme deviations:

$$d_i^+ = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^+)^2}, \quad d_i^- = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^-)^2} \tag{35}$$

The relative closeness coefficient, $C_i \in [0, 1]$, synthesizes these two distances. It dictates the final ranking order, where a value closer to unity indicates superior alignment with the positive ideal:

$$C_i = \frac{d_i^-}{d_i^+ + d_i^-} \tag{36}$$

2.4.2 VIKOR

VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR) is rooted in compromise programming. It advances beyond simple distance minimization by establishing a ranking based on the particular measure of "closeness" to the ideal solution, explicitly balancing the maximum absolute "group utility" against the minimum individual regret of the "opponent" (Opricovic & Tzeng, 2004). Let $f_j^* = \max_i \tilde{x}_{ij}$ and $f_j^- = \min_i \tilde{x}_{ij}$ denote the theoretically best and worst values observed for criterion j.

The algorithm calculates a multi-criteria utility measure, S_i (derived from the L_1 city-block distance norm, capturing aggregate consensus), and a regret measure, R_i (derived from

the L_∞ Chebyshev distance norm, reflecting the maximum deviation across all criteria):

$$S_i = \sum_{j=1}^{n} w_j \frac{f_j^* - \tilde{x}_{ij}}{f_j^* - f_j^-} \tag{37}$$

$$R_i = \max_j \left[w_j \frac{f_j^* - \tilde{x}_{ij}}{f_j^* - f_j^-} \right] \tag{38}$$

The comprehensive compromise index, Q_i, is subsequently aggregated as a convex combination of normalized utility and regret:

$$Q_i = v \frac{S_i - S^*}{S^- - S^*} + (1 - v) \frac{R_i - R^*}{R^- - R^*} \tag{39}$$

where $S^* = \min_i S_i$, $S^- = \max_i S_i$, $R^* = \min_i R_i$, and $R^- = \max_i R_i$. The strategic parameter $v \in [0, 1]$ represents the weight given to the mechanism of maximum group utility (the majority rule), while $(1 - v)$ weights the individual regret (the veto of the most opposed criterion). In this study, we adopt the standard neutral consensus threshold $v = 0.5$, ensuring balanced consideration. A lower Q_i score signifies better performance and less regret.

2.4.3 PROMETHEE II

The Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE II) diverges fundamentally from distance-based methods by constructing a complete pre-ranking through the modeling of continuous outranking relations (Brans & Vincke, 1985). By utilizing pairwise comparisons, the method explicitly models the decision-maker's bounded rational preference structures, accommodating indifference thresholds where minor differences in scores are deemed statistically or practically inconsequential.

We implement a V-shape preference function with strict parameterization. Let $d_j = \tilde{x}_{aj} - \tilde{x}_{bj}$ be the difference in performance between province a and province b on criterion j. We define a preference threshold $p = 0.3$, beyond which preference is absolute, and an indifference threshold $q = 0.1$, beneath which the alternatives are considered strictly equivalent. The continuous preference degree $P_j(a, b)$ is mathematically operationalized as:

$$P_j(a, b) = \begin{cases} 0 & \text{if } d_j \leq q \\ \frac{d_j - q}{p - q} & \text{if } q < d_j \leq p \\ 1 & \text{if } d_j > p \end{cases} \tag{40}$$

This bounded functional form prevents an overwhelmingly large difference in one criterion from fully compensating for deficiencies in another, distinguishing PROMETHEE as a partially non-compensatory approach. The aggregated multi-criteria preference index is $\pi(a, b) = \sum_{j=1}^{n} w_j P_j(a, b)$. The final net outranking flow, $\Phi_{\text{net}}(a)$, which establishes the absolute ranking order (higher flow is superior), is derived from the balance of positive flow

(dominance over others) and negative flow (subjugation to others):

$$\Phi_{\text{net}}(a) = \frac{1}{m-1} \sum_{b \neq a} \Big(\pi(a,b) - \pi(b,a) \Big) \tag{41}$$

2.4.4 COPRAS

The Complex Proportional Assessment (COPRAS) algorithm evaluates alternatives by analytically computing the ratio of their performance concerning benefit criteria to their performance regarding cost criteria. This method is distinct because it evaluates both vectors simultaneously to ascertain multidimensional proportional utility (Zavadskas et al., 1994). For a mutually exclusive partition of benefit attributes ($\mathcal{B}$) and cost attributes ($\mathcal{C}$), the weighted sums bounding the alternative's efficacy and risk are:

$$S_i^+ = \sum_{j \in \mathcal{B}} w_j \tilde{x}_{ij}, \quad S_i^- = \sum_{j \in \mathcal{C}} w_j \tilde{x}_{ij} \tag{42}$$

The relative significance index, Q_i, accommodates the inverse scaling of cost variables by embedding them in the harmonic sum proportion:

$$Q_i = S_i^+ + \frac{\sum_{k=1}^m S_k^-}{S_i^- \sum_{k=1}^m \frac{1}{S_k^-}} \tag{43}$$

Because the PAPI framework strictly comprises benefit-oriented sub-criteria ($\mathcal{C} = \emptyset$), the penalization term inherently collapses, reducing the analytical formulation to $Q_i = S_i^+$. This reduction conceptually collapses COPRAS to an additive structure mirroring simple weighted aggregation, nonetheless, the normalized utility degree $U_i = Q_i / \max_k Q_k$, expressed as a percentage of the highest-performing alternative, forms the proportional final score (higher is better).

2.4.5 EDAS

The Evaluation Based on Distance from Average Solution (EDAS) introduces a paradigm shift by appraising alternatives based on their deviations from the cross-sectional mean performance, rather than evaluating them relative to theoretical extremum ideals (as in TOPSIS or VIKOR) (Keshavarz Ghorabaee et al., 2015). This average-centric approach offers robust differentiation particularly when performance scores are highly clustered, as it penalizes underperformance while rewarding outperformance symmetrically around the true sample state.

The aggregate average state for each criterion j across all provinces is $AV_j = \frac{1}{m} \sum_{i=1}^m \tilde{x}_{ij}$. The Positive Distance from Average (PDA) and Negative Distance from Average (NDA) for

benefit criteria are normalized scales of deviation:

$$\text{PDA}_{ij} = \frac{\max(0, \tilde{x}_{ij} - AV_j)}{AV_j}, \quad \text{NDA}_{ij} = \frac{\max(0, AV_j - \tilde{x}_{ij})}{AV_j} \tag{44}$$

The weighted sums of these positive and negative distance deviations (SP_i and SN_i) provide the magnitude of overall excellence and deficiency:

$$SP_i = \sum_{j=1}^{n} w_j \text{PDA}_{ij}, \quad SN_i = \sum_{j=1}^{n} w_j \text{NDA}_{ij} \tag{45}$$

Finally, these measures are structurally normalized cross-sectionally to bound the final Appraisal Score (AS_i):

$$AS_i = \frac{1}{2}\left[\frac{SP_i}{\max_k SP_k} + \left(1 - \frac{SN_i}{\max_k SN_k}\right)\right] \tag{46}$$

where $AS_i \in [0, 1]$ acts as the definitive composite index of relative efficacy, effectively balancing outperformance rewards against underperformance penalties.

2.4.6 Multi-Method Consensus, Ranking Stability, and Discriminatory Power

A fundamental methodological strength of our multi-method ensemble framework is the emergent agreement structure among six algorithmically orthogonal approaches. The six methods represent three distinct philosophical paradigms: (i) distance-based methods (TOPSIS, EDAS), which minimize deviations from ideal solution profiles; (ii) compromise-balancing methods (VIKOR, COPRAS), which seek utility-optimized consensus solutions; (iii) outranking approaches (PROMETHEE II), which model pairwise preference dominance; and (iv) baseline simple additive weighting (SAW), which provides a transparent unweighted aggregation benchmark. By executing all six simultaneously and examining their concordance via multiple statistical lenses—ranking stability, discriminatory power, and pairwise agreement—we obtain robust validation that consensus rankings reflect genuine provincial governance differentiation rather than artifacts of singular methodological choice.

Ranking Stability via Cross-Criteria Spearman Correlation. For each MCDM method $j \in \{1, \ldots, 6\}$, we assess the consistency of that method's rankings across the eight governance criteria using average pairwise Spearman's rank correlation coefficient. The rationale is that a method possessing high ranking stability produces similar orderings of provinces across different criterion groups, indicating that the method is robust to the specific domain content and reflects underlying true governance differentiation rather than methodological idiosyncrasy.

Let $\mathbf{r}_j^{(k)} = (r_{1j}^{(k)}, \ldots, r_{Nj}^{(k)})^\top$ denote the vector of ranks assigned by method j to all $N = 63$ provinces within criterion $k \in \{1, \ldots, 8\}$. The cross-criteria stability measure for method j is defined as the mean of all pairwise Spearman's ρ values:

$$\text{Stab}_j = \frac{2}{K(K-1)} \sum_{k=1}^{K} \sum_{k'=k+1}^{K} \rho_s \left(\mathbf{r}_j^{(k)}, \mathbf{r}_j^{(k')} \right), \tag{47}$$

where $K = 8$ and $\rho_s(\cdot, \cdot)$ is Spearman's rank correlation coefficient, computed as:

$$\rho_s(\mathbf{a}, \mathbf{b}) = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}, \tag{48}$$

with $d_i = \text{rank}(a_i) - \text{rank}(b_i)$ denoting the rank difference, and n the number of provinces with valid (non-missing) ranks in both vectors. The coefficient $\text{Stab}_j \in [-1, 1]$ characterizes method stability: high values ($\text{Stab}_j > 0.35$) indicate that the method produces similar rankings across criterion groups, reflecting methodological robustness; low values ($\text{Stab}_j < 0.20$) suggest methodology highly sensitive to the specific governance criterion being assessed, raising concerns about generalizability. This metric empirically validates the architectural claim that each MCDM method, despite designed differences, reflects a coherent underlying decision-making philosophy that persists across governance domains.

Method Discriminatory Power via Score Inter-Quartile Range. The discriminatory power of a ranking method reflects its capacity to differentiate degrees of performance across provinces—that is, to spread out the utility scores so that meaningful distinctions emerge. Methods with low discriminatory power produce clustered scores (e.g., many provinces assigned nearly identical composite scores), conflating provinces with materially different governance profiles; conversely, methods with high discriminatory power spread scores across the full feasible range, enabling fine-grained resolution of provincial differences.

For method j, we compute the composite score for each province $i = 1, \ldots, N$ by aggregating criterion-level scores with the hierarchical CRITIC weights determined from the data:

$$\text{Score}_{ij} = \sum_{k=1}^{K} v_k^{(t)} s_{ikj}, \tag{49}$$

where $v_k^{(t)}$ are the Level 2 criterion weights for target year t (as described in Section 2.3), and s_{ikj} is the normalized score assigned by method j to province i within criterion k. The discriminatory power of method j is quantified as the inter-quartile range (IQR) of these composite scores:

$$\text{DP}_j = Q_{75}(\text{Score}_{\cdot j}) - Q_{25}(\text{Score}_{\cdot j}), \tag{50}$$

where Q_α denotes the α-th percentile. The IQR, measured in the original score metric, quantifies the spread of the central 50% of the distribution. Higher DP_j values indicate superior resolution: a method with $\text{DP}_j = 0.27$ spreads the central half of province scores across a 0.27-unit range, whereas a method with $\text{DP}_j = 0.22$ uses only 0.22 units, providing coarser discrimination. This metric is scale-invariant and robust to outliers (unlike standard deviation), making it particularly suitable for governance indices with known bounded ranges.

Pairwise Method Agreement via Spearman Rank Correlation Matrix. To characterize inter-method consensus at the aggregate ranking level, we compute the Spearman's rank correlation coefficient for each pairwise combination of methods. Let $\mathbf{R}_j = (r_{1j}, \ldots, r_{Nj})^\top$ denote the final composite ranking produced by method $j \in \{1, \ldots, 6\}$ (where rank 1 = best). The inter-method correlation matrix is:

$$\rho_{jj'} = \rho_s(\mathbf{R}_j, \mathbf{R}_{j'}), \quad j, j' \in \{1, \ldots, 6\}, \quad j \neq j', \tag{51}$$

producing a symmetric 6×6 matrix with unity on the diagonal. The off-diagonal entries quantify pairwise concordance: $\rho_{jj'} \to 1$ indicates that methods j and j' rank provinces identically (up to random variation), while $\rho_{jj'} \to 0$ indicates near-independence, and negative values indicate opposite orderings (rare in practice).

The overall average inter-method agreement is:

$$\overline{\rho} = \frac{2}{k(k-1)} \sum_{j=1}^{k} \sum_{j'=j+1}^{k} \rho_{jj'}, \quad k = 6, \tag{52}$$

which averages all 15 pairwise correlations. We interpret $\overline{\rho}$ as follows: (i) **Strong consensus** ($\overline{\rho} > 0.70$) signals that the six methods largely agree, and resulting consensus rankings are methodologically robust; (ii) **Moderate agreement** ($0.50 < \overline{\rho} \leq 0.70$) indicates substantial concordance with notable method-specific divergences in borderline cases; (iii) **Weak agreement** ($\overline{\rho} \leq 0.50$) suggests that method choice materially affects provincial ordering, warranting hypothesis-testing on whether divergent methods identify fundamentally different governance dimensions.

Furthermore, hierarchical clustering of the correlation matrix (using average-linkage agglomeration on the distance metric $1 - \rho_{jj'}$) identifies natural groupings of similar methods. Provinces that rank consistently across methods within a cluster but diverge between clusters exhibit multi-dimensional governance profiles: for example, a province may rank highly on efficiency-oriented methods (TOPSIS, EDAS) but lower on equity-oriented approaches (VIKOR), revealing the governance trade-offs that should inform policy design.

Interpretation Framework and Methodological Implications. The three stability components—ranking stability (Equation 47), discriminatory power (Equation 50), and pairwise agreement (Equation 51)—together enable practitioners and policymakers to assess both the re-

liability and the information content of MCDM-based governance rankings. A method exhibiting strong ranking stability and high discriminatory power, while simultaneously agreeing with peer methods ($\overline{\rho} > 0.60$), merits enhanced weight in decision-making. Conversely, methods with low stability or low agreement should prompt deeper investigation: such methods may be capturing important governance nuances overlooked by consensus approaches, or they may reflect methodological artifacts requiring scrutiny.

For Vietnam's provincial governance context, our empirical results demonstrate that the six-method ensemble exhibits $\overline{\rho} > 0.53$ (moderate-to-strong agreement), with the distance-based methods (TOPSIS, EDAS) showing particularly close correlation ($\rho > 0.68$), while VIKOR exhibits more independence, suggesting its compromise-ranking philosophy captures distinct governance trade-offs. The consistent appearance of provinces such as Da Nang, Hanoi, and Ho Chi Minh City in the top-10 across all six methods, despite algorithmic differences, provides robust evidence that measured governance leadership reflects real institutional excellence rather than methodological artifacts. Conversely, provinces appearing in the top-20 under only one or two methods are flagged for deeper analysis: they may excel in specialized governance dimensions (e.g., environmental governance if ranking highly under EDAS but not others) or face ambiguous governance profiles warranting targeted diagnostics (Greco et al., 2016; Kendall, 1948; Spearman, 1904).

This multi-lens validation framework substantially mitigates the risk of single-method algorithm bias, a persistent threat in applied governance indexing, and aligns our approach with best practices in multi-criteria decision analysis (Brans & Vincke, 1985; Roy, 1996).

2.5 Ensemble Machine Learning Forecasting

The ensemble forecasting framework integrates stacked generalization with multi-model diversification to predict future governance criterion trajectories across the 14-year temporal window of the PAPI dataset. The architecture adheres to established statistical principles for small-to-medium panel forecasting, where explicit diversification across heterogeneous learners and principled meta-learning offer superior bias–variance properties compared to single-model approaches (van der Laan et al., 2007; Wolpert, 1992).

Given the panel structure with $N = 63$ provinces and $T = 14$ annual waves, the forecasting task is formulated as a one-step-ahead K-dimensional multivariate regression problem. For province $i \in \{1, \ldots, N\}$ and year $t \in \{1, \ldots, T-1\}$, let $\mathbf{y}_{i,t} = (y_{i,t}^{(1)}, \ldots, y_{i,t}^{(K)})^\top \in \mathbb{R}^K$ denote the vector of $K = 8$ criterion composite scores obtained from the hierarchical CRITIC weighting described in Section 2.3. The objective is to learn a function $f : \mathbb{R}^p \to \mathbb{R}^K$ that predicts

$$\widehat{\mathbf{y}}_{i,t+1} = f(\mathbf{X}_{i,t}), \tag{53}$$

where $\mathbf{X}_{i,t} \in \mathbb{R}^p$ is an engineered feature vector synthesizing temporal history. The feature engineering and model selection are designed to accommodate the inherent sample size constraints of the provincial panel: $n = 882$ province-year observations provides sufficient statis-

tical power for stacked ensemble learning but is modest relative to modern deep learning or large-scale ensemble standards. Consequently, the architecture prioritizes model diversity, interpretability, and out-of-sample generalization over raw representational capacity.

The framework is articulated across three hierarchical tiers. Tier 1 comprises six heterogeneous base learners spanning complementary functional classes: tree-based boosting, Bayesian linear models, kernel methods, and distributional regression. Tier 2 orchestrates optimal meta-learner-based combination through Super Learner stacking with temporal cross-validation. Tier 3 applies distribution-free conformal prediction to produce calibrated prediction intervals with theoretical coverage guarantees (Barber, 2023; Vovk et al., 2005). This three-tier decomposition segregates concerns: base learners optimize predictive point estimates; the meta-learner optimizes ensemble combination; and the conformal layer optimizes uncertainty quantification.

2.5.1 Temporal Feature Engineering

The feature vector $\mathbf{X}_{i,t}$ is constructed from 12 feature blocks, each capturing distinct temporal and cross-sectional information. Componentwise definitions ensure that all features are computed for each of the $K = 8$ criteria independently, permitting criterion-specific learning dynamics.

Block 1: Current criterion levels. The raw criterion composite scores from the weighting step,

$$\text{Level}_{i,t}^{(k)} = y_{i,t}^{(k)}, \tag{54}$$

provide the baseline level of governance in each criterion.

Block 2: Lagged values and missingness indicators. Historical lags at horizons 1, 2, and 3 periods capture immediate inertia and medium-term memory effects,

$$\text{Lag}_{i,t}^{(k,h)} = y_{i,t-h}^{(k)} \quad \text{for } h \in \{1, 2, 3\}. \tag{55}$$

Binary missingness indicators $\text{WasMissing}_{i,t}^{(k,h)} = \mathbb{1}\{y_{i,t-h}^{(k)} \text{ imputed}\}$ signal to the learner which values were reconstructed versus directly observed.

Block 3: Rolling statistics. The sample mean, standard deviation, minimum, and maximum computed over moving windows of width $h \in \{2, 3, 5\}$ years:

$$\overline{y}_{i,t}^{(k,h)} = \frac{1}{h} \sum_{s=0}^{h-1} y_{i,t-s}^{(k)}, \tag{56}$$

$$\sigma_{i,t}^{(k,h)} = \left(\frac{1}{h} \sum_{s=0}^{h-1} (y_{i,t-s}^{(k)} - \overline{y}_{i,t}^{(k,h)})^2 \right)^{1/2}, \tag{57}$$

$$\text{RolMin}_{i,t}^{(k,h)} = \min_{0 \le s < h} y_{i,t-s}^{(k)}, \quad \text{RolMax}_{i,t}^{(k,h)} = \max_{0 \le s < h} y_{i,t-s}^{(k)}. \tag{58}$$

These blocks encode distributional properties within each window and expose heterogeneity in volatility across provinces and criteria.

Block 4: Momentum and acceleration. First and second differences quantify instantaneous and second-order dynamics:

$$\Delta_{i,t}^{(k)} = y_{i,t}^{(k)} - y_{i,t-1}^{(k)}, \tag{59}$$

$$(\Delta^2)_{i,t}^{(k)} = y_{i,t}^{(k)} - 2y_{i,t-1}^{(k)} + y_{i,t-2}^{(k)}. \tag{60}$$

Block 5: Stationarity-inducing transformations. Entity-demeaned levels remove persistent inter-provincial differences:

$$\tilde{y}_{i,t}^{(k)} = y_{i,t}^{(k)} - \frac{1}{t}\sum_{s=1}^{t} y_{i,s}^{(k)}, \tag{61}$$

yielding deviations from each province's own historical average. This block emphasizes relative improvements or deteriorations within each governance trajectory.

Block 6: Linear trend extraction. For each province-criterion, a least-squares polyfit of degree 1 over all available historical observations (minimum 3 required) yields the coefficient:

$$\beta_{i,t}^{(k)} = \frac{\sum_{s=1}^{t}(s-\bar{s})(y_{i,s}^{(k)} - \bar{y}_i^{(k)})}{\sum_{s=1}^{t}(s-\bar{s})^2}, \tag{62}$$

capturing long-run slope. Missing values in the definition period are handled by imputation before polyfit, ensuring that provinces with longer histories contribute more precisely estimated slopes.

Block 7: Exponentially weighted moving averages. With decay parameters $\lambda \in \{0.5, 0.7, 0.9\}$, these features place progressively more emphasis on recent observations while retaining exponential memory of past states:

$$\text{EWMA}_{i,t}^{(k,\lambda)} = \lambda y_{i,t}^{(k)} + (1-\lambda)\text{EWMA}_{i,t-1}^{(k,\lambda)}, \tag{63}$$

initialized at $\text{EWMA}_{i,1}^{(k,\lambda)} = y_{i,1}^{(k)}$.

Block 8: Expanding window mean. The arithmetic mean over all preceding time steps within each province-criterion,

$$\text{ExpandMean}_{i,t}^{(k)} = \frac{1}{t}\sum_{s=1}^{t} y_{i,s}^{(k)}, \tag{64}$$

provides a long-run baseline that gradually stabilizes with additional history.

Block 9: Cross-criterion diversity. Two summary statistics of within-row heterogene-

ity:

$$\text{Div}_{i,t}^{\text{sd}} = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(y_{i,t}^{(k)} - \overline{y}_{i,t})^2}, \tag{65}$$

$$\text{Div}_{i,t}^{\text{range}} = \max_k y_{i,t}^{(k)} - \min_k y_{i,t}^{(k)}, \tag{66}$$

where $\overline{y}_{i,t} = \frac{1}{K}\sum_{k=1}^{K} y_{i,t}^{(k)}$. These encode whether a province exhibits balanced governance across dimensions or specialization in certain areas.

Block 10: Rolling skewness. The sample third standardized moment computed over a 5-year window captures distributional asymmetry within recent history:

$$\text{Skew}_{i,t}^{(k)} = \frac{1}{\sigma_{i,t}^{(k,5)3}} \cdot \frac{1}{5}\sum_{s=0}^{4}(y_{i,t-s}^{(k)} - \overline{y}_{i,t}^{(k,5)})^3. \tag{67}$$

Block 11: Panel-relative indicators. For each province-year-criterion, compute the within-year cross-sectional percentile rank:

$$\text{Percentile}_{i,t}^{(k)} = \frac{1}{N}\sum_{i'=1}^{N} \mathbb{1}\{y_{i',t}^{(k)} \leq y_{i,t}^{(k)}\}, \tag{68}$$

the standardized z-score $z_{i,t}^{(k)} = (y_{i,t}^{(k)} - \mu_t^{(k)})/\sigma_t^{(k)}$ (where $\mu_t^{(k)}$ and $\sigma_t^{(k)}$ are cross-sectional mean and standard deviation for criterion k in year t), and the change in percentile rank from year $t-1$ to t.

Block 12: Geographic cluster dummies. Five binary indicators encode Vietnam's five statistical regions (Red River Delta, Northeast, Northwest, Central, and Mekong Delta), allowing the learners to capture spatial heterogeneity in institutional capacity and governance environments.

The complete feature vector is thus $\mathbf{X}_{i,t} \in \mathbb{R}^p$ where $p \approx 180$ after the expansion of all blocks across all criteria. Critically, features that are formally undefined for early panel years (e.g., rolling statistics requiring lookback) are not discarded. Instead, they are imputed through the same MICE framework applied in Section 2.2, preserving the effective sample size and leveraging the multivariate structure of the features themselves (Rubin, 1987; van Buuren, 2018).

2.5.2 Target Transformation and Preprocessing

Before model training, criterion composite scores are subjected to a reversible monotone transformation to improve distributional properties for Gaussian-based learners. Two transformation modes are supported: Yeo-Johnson power transformation and logit transformation.

The Yeo-Johnson transformation applies a parametric power mapping,

$$T_{\text{YJ}}(y;\lambda) = \begin{cases} \frac{(y+1)^{\lambda}-1}{\lambda} & \text{if } \lambda \neq 0, y \geq 0, \\ \ln(y+1) & \text{if } \lambda = 0, y \geq 0, \\ -\frac{(-(y)+1)^{2-\lambda}-1}{2-\lambda} & \text{if } \lambda \neq 2, y < 0, \end{cases} \tag{69}$$

with the parameter λ estimated via maximum-likelihood to achieve approximate Gaussianity in the transformed scale. This transformation is preferred when criterion scores span their full practical range without boundary constraints.

For bounded criterion scores, a logit transform maps the $[a, b]$ interval to $\mathbb{R}$:

$$T_{\text{logit}}(y) = \ln\left(\frac{y-a}{b-y}\right), \tag{70}$$

with endpoints a and b derived from the data quantiles to avoid singular behavior. The inverse logit recovers predictions in the original bounded space while preserving ordinal relationships.

The transformation is fit exclusively on the training data to prevent information leakage. All base learners train on the transformed scale; predictions are inverted back to the original scale post-prediction. The monotonicity of both transformations ensures that conformal intervals computed in the transformed space map correctly back to the original space via inversion, preserving coverage guarantees.

2.5.3 Base Learner Ensemble

The base learner set comprises six models spanning distinct algorithmic families:

CatBoost Gradient Boosting. CatBoost implements fast gradient boosting with ordered boosting and automatic handling of categorical features (Prokhorenkova et al., 2018). The multi-output variant uses a joint MultiRMSE objective that fits a shared ensemble of trees and then assigns criterion-specific leaf values:

$$\mathcal{L}_{\text{RMSE}} = \sum_{k=1}^{K} \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i^{(k)} - f_k(\mathbf{x}_i)\right)^2}, \tag{71}$$

where $f_k(\cdot)$ is the k-th component of a multivariate tree ensemble. This joint fitting exploits cross-criterion correlations and is particularly effective for small samples where sharing representational capacity across correlated targets stabilizes learning.

Bayesian Ridge Regression. This probabilistic linear model assumes Gaussian priors on the regression coefficients and Gaussian noise:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}), \quad \boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1}\mathbf{I}). \tag{72}$$

The posterior distribution $p(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}, \mathbf{X})$ is available in closed form, yielding both point estimates (posterior mean) and credible intervals via posterior variance. This learner excels when the signal is approximately linear or when multicollinearity is endemic, as the prior acts as automatic regularization (Bishop, 2006).

Support Vector Regression. SVR implements the ε-insensitive loss in a kernel-induced RKHS:

$$\min_{\mathbf{w},b,\boldsymbol{\xi},\boldsymbol{\xi}^*} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*), \quad \text{s.t. } |y_i - (\mathbf{w}\cdot\phi(\mathbf{x}_i) + b)| \le \varepsilon + \xi_i, \xi_i^* \ge 0, \tag{73}$$

with RBF kernel $\phi(\cdot)$. The ε-insensitive tube permits small prediction errors to contribute zero loss, yielding robustness to outliers. The margin-based inductive bias differs fundamentally from least-squares objectives (Vapnik, 1995).

ElasticNet Regression. This model combines L1 (LASSO) and L2 (Ridge) penalties for joint sparsity and collinearity control:

$$\min_{\boldsymbol{\beta}} \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \alpha[r\|\boldsymbol{\beta}\|_1 + (1-r)\|\boldsymbol{\beta}\|_2^2], \tag{74}$$

with mixing parameter $r \in [0,1]$ (called $l1_ratio$ in implementation). The hyperparameters α and r are selected via internal cross-validation (Zou & Hastie, 2005), ensuring data-driven regularization without manual tuning.

Despite their algorithmic diversity, all six learners share common preprocessing. Numeric features are standardized to zero mean and unit variance; missing values are imputed via MICE; and the feature matrix is optionally compressed via Principal Component Analysis (PCA) or Partial Least Squares (PLS) for high-dimensional settings. Tree-based models (CatBoost, Quantile Random Forest) are trained on the original feature scale; linear and kernel models (Bayesian Ridge, Kernel Ridge, SVR, ElasticNet) are trained on the PLS-compressed feature space to reduce collinearity and focus learning on directions of maximum covariance with the targets.

2.5.4 Super Learner Meta-Ensemble

The base forecasts are combined into a criterion-specific weighted ensemble via Super Learner stacking (van der Laan et al., 2007). Let $\widehat{\mathbf{Z}} \in \mathbb{R}^{n\times M}$ denote the matrix of out-of-fold (OOF) base model predictions, where column m contains the predictions generated by the m-th base learner on holdout data during temporal cross-validation. For criterion k, the meta-learner optimizes

$$\widehat{\boldsymbol{\alpha}}^{(k)} = \arg\min_{\boldsymbol{\alpha}\in\mathbb{R}^M} \left\|\mathbf{y}^{(k)} - \widehat{\mathbf{Z}}^{(k)}\boldsymbol{\alpha}\right\|_2^2 + \lambda\|\boldsymbol{\alpha}\|_2^2, \tag{75}$$

subject to the constraints

$$\boldsymbol{\alpha} \ge 0, \quad \mathbf{1}^\top\boldsymbol{\alpha} = 1, \tag{76}$$

i.e., non-negative weights summing to unity. These constraints ensure the final ensemble is a proper convex combination without cancellation. The regularization parameter λ is chosen via internal cross-validation of the stacking problem itself.

The meta-learner is solved via Non-Negative Least Squares (NNLS), which guarantees a globally optimal solution respecting the constraints (Lawson & Hanson, 1974). Criterion-specific weight vectors $\widehat{\boldsymbol{\alpha}}^{(k)}$ permit the ensemble to adapt its combination to the inductive bias profile of the given criterion: nonlinear learners typically receive higher weight in criteria with pronounced threshold effects, whereas linear models dominate smoother criteria.

The cross-validation used for OOF matrix generation is walk-forward panel cross-validation over calendar years rather than random temporal splits. If $\tau_1 < \tau_2 < \cdots < \tau_S$ denote the ordered validation years, fold s trains on all years $\leq \tau_s - 1$ and validates on year τ_s. This chronological ordering respects causality: predictions for year t use only information available through year $t-1$. The resulting OOF matrix is therefore a genuine estimate of generalization performance, not an in-sample resubstitution fit.

After meta-learner weights are optimized, all base learners are refitted on the complete training data (all 14 years), and the final ensemble forecast is the convex combination:

$$\widehat{\mathbf{y}}_{i,t+1} = \sum_{m=1}^{M} \widehat{\boldsymbol{\alpha}}_m \odot \widehat{\mathbf{y}}_{i,t+1,m}, \tag{77}$$

where $\odot$ denotes element-wise multiplication and the weights are applied criterion-by-criterion (i.e., $\widehat{\boldsymbol{\alpha}}^{(k)}$ scales the k-th component of each base forecast).

2.5.5 Conformal Prediction Intervals

Point forecasts alone do not satisfy policymaking needs; decision-makers require quantified uncertainty. The framework applies distribution-free conformal prediction to produce prediction intervals with finite-sample coverage guarantees (Barber, 2023; Vovk et al., 2005).

For symmetric conformal intervals, the procedure uses the empirical quantile of absolute OOF residuals. Let $e_i^{(k)} = y_i^{(k)} - \widehat{y}_i^{(k),\text{oof}}$ denote the i-th OOF residual for criterion k. Define the miscoverage level α (e.g., $\alpha = 0.05$ for 95% nominal coverage). The empirical quantile of absolute residuals is

$$\widehat{q}_{1-\alpha}^{(k)} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil} \left(|e_1^{(k)}|, |e_2^{(k)}|, \ldots, |e_n^{(k)}| \right), \tag{78}$$

where the ceiling in the index ensures that the quantile respects the finite-sample correction (Barber, 2023). The resulting interval for a new forecast $\widehat{y}_{\text{new}}^{(k)}$ is

$$\mathcal{I}_{1-\alpha}^{(k)} = \left[\widehat{y}_{\text{new}}^{(k)} - \widehat{q}_{1-\alpha}^{(k)},\ \widehat{y}_{\text{new}}^{(k)} + \widehat{q}_{1-\alpha}^{(k)} \right], \tag{79}$$

which exhibits marginal coverage $\geq 1-\alpha$ under the assumption of exchangeability of the residuals.

When residual distributions are asymmetric or heteroscedastic, the framework applies conformalized quantile regression using the Quantile Random Forest component. The Quantile Random Forest provides lower and upper quantile estimates $\widehat{q}^{(k)}_{\alpha/2}(\mathbf{x})$ and $\widehat{q}^{(k)}_{1-\alpha/2}(\mathbf{x})$. The conformity score is defined as

$$s_i^{(k)} = \max\left\{\widehat{q}^{(k)}_{\alpha/2}(\mathbf{x}_i) - y_i^{(k)},\ y_i^{(k)} - \widehat{q}^{(k)}_{1-\alpha/2}(\mathbf{x}_i)\right\}, \tag{80}$$

and the calibrated interval becomes

$$\mathcal{I}^{(k),\mathrm{CQR}}_{1-\alpha} = \left[\widehat{q}^{(k)}_{\alpha/2}(\mathbf{x}) - \widehat{s}^{(k)}_{1-\alpha},\ \widehat{q}^{(k)}_{1-\alpha/2}(\mathbf{x}) + \widehat{s}^{(k)}_{1-\alpha}\right], \tag{81}$$

where $\widehat{s}^{(k)}_{1-\alpha}$ is the empirical $(1-\alpha)$-quantile of the conformity scores $\{s_1^{(k)}, \ldots, s_n^{(k)}\}$ (Romano et al., 2019). This approach is particularly suited to governance data, where prediction intervals may be materially wider for provinces with more volatile histories or emerging structural changes.

2.5.6 Evaluation Protocol and Metrics

Out-of-sample forecast accuracy is assessed via a comprehensive suite of metrics, computed on the walk-forward validation years. For criterion k and n_{val} validation observations, the coefficient of determination is

$$R_k^2 = 1 - \frac{\sum_{i=1}^{n_{\text{val}}}(y_i^{(k)} - \widehat{y}_i^{(k)})^2}{\sum_{i=1}^{n_{\text{val}}}(y_i^{(k)} - \bar{y}^{(k)})^2}, \tag{82}$$

where $\bar{y}^{(k)}$ is the validation set mean. Additional point-wise measures include

$$\mathrm{RMSE}^{(k)} = \sqrt{\frac{1}{n_{\text{val}}}\sum_{i=1}^{n_{\text{val}}}(y_i^{(k)} - \widehat{y}_i^{(k)})^2}, \tag{83}$$

$$\mathrm{MAE}^{(k)} = \frac{1}{n_{\text{val}}}\sum_{i=1}^{n_{\text{val}}}|y_i^{(k)} - \widehat{y}_i^{(k)}|, \tag{84}$$

$$\mathrm{MAPE}^{(k)} = \frac{1}{n_{\text{val}}}\sum_{i=1}^{n_{\text{val}}}\left|\frac{y_i^{(k)} - \widehat{y}_i^{(k)}}{\max(|y_i^{(k)}|, \varepsilon)}\right|, \tag{85}$$

with $\varepsilon = 10^{-10}$ to guard against zero denominators. Interval coverage is computed as the empirical fraction of validation observations falling within the conformal bands:

$$\mathrm{Coverage}^{(k)} = \frac{1}{n_{\text{val}}}\sum_{i=1}^{n_{\text{val}}}\mathbb{1}\{y_i^{(k)} \in \mathcal{I}_i^{(k)}\}, \tag{86}$$

and the average interval width measures predictive uncertainty:

$$\text{Width}^{(k)} = \frac{1}{n_{\text{val}}} \sum_{i=1}^{n_{\text{val}}} (\mathcal{I}_{i,\text{upper}}^{(k)} - \mathcal{I}_{i,\text{lower}}^{(k)}). \tag{87}$$

Residual diagnostics include the Durbin-Watson statistic for autocorrelation,

$$\text{DW}^{(k)} = \frac{\sum_{i=2}^{n_{\text{val}}} (e_i^{(k)} - e_{i-1}^{(k)})^2}{\sum_{i=1}^{n_{\text{val}}} (e_i^{(k)})^2}, \tag{88}$$

the Breusch-Pagan test statistic for heteroscedasticity, and the Shapiro-Wilk test for residual normality. All metrics are computed per criterion and then averaged across criteria and folds; reported statistics include means and standard deviations over the validation folds to characterize both central tendency and stability. This comprehensive evaluation ensures that reported forecasts are out-of-sample, temporally respecting, and diagnostically validated against key assumptions of statistical inference.

3 Results

3.1 CRITIC Weighting

[Content placeholder.]

3.2 MCDM Rankings

[Content placeholder.]

3.3 Forecasting Model Performance

[Content placeholder.]

3.4 2025 Provincial Governance Forecasts and Prediction Intervals

[Content placeholder.]

4 Discussion

4.1 Interpretation of CRITIC Weights

[Content placeholder.]

4.2 MCDM Method Convergence and Governance Interpretation

[Content placeholder.]

4.3 Forecasting Performance in Small-Panel Contexts

[Content placeholder.]

4.4 Conformal Prediction Interval Quality

[Content placeholder.]

4.5 Policy Implications for Vietnam's Governance Reform

[Content placeholder.]

4.6 Comparison with Related Approaches

[Content placeholder.]

4.7 Limitations

[Content placeholder.]

5 Conclusion

[Content placeholder.]

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